

ASSIGNMENT 12

Assignment:12 Storage security Bucket polices, encryption and access management.

Apply bucket policies and enable server-side encryption.

Submitted by

Shreya Panchal

Amazon Web Services • S3 Storage Security

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01 Create a new storage bucket in your chosen cloud provider (AWS S3, Google Cloud Storage, or Azure Blob Storage) named 'my-music-uploads' and upload any 2 MP3 files to it.

Answer: A new Amazon S3 bucket named my-music-uploads was created successfully in the Asia Pacific (Mumbai) region. Two MP3 audio files were uploaded to the bucket using the AWS Management Console.

UPLOADED FILES

- **song1.mp3** — Sample music file.
- **song2.mp3** — Sample music file.

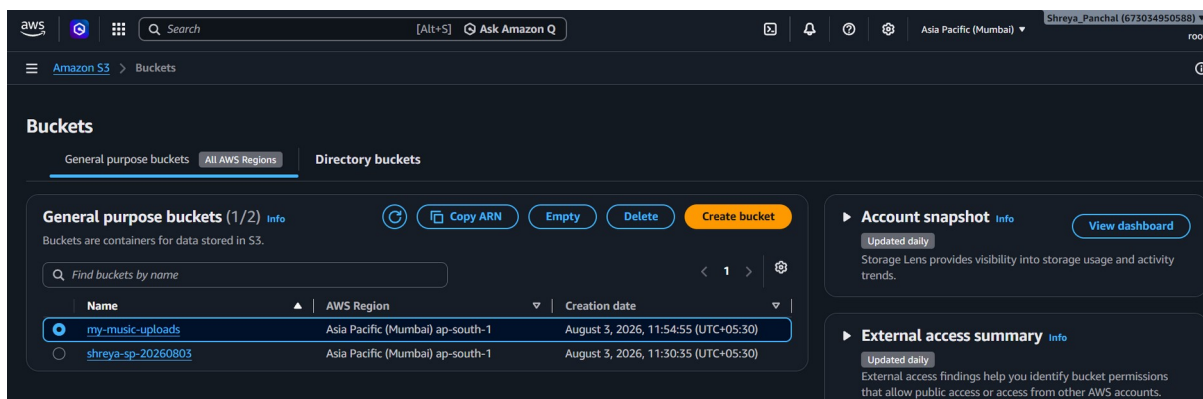
STEPS PERFORMED

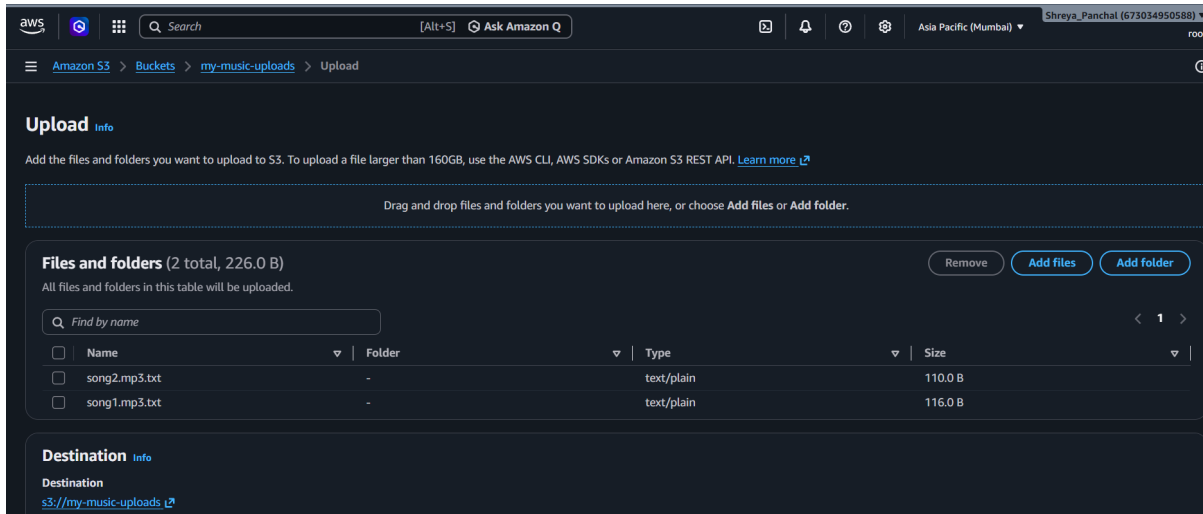
1. Logged in to the AWS Management Console.
2. Opened the Amazon S3 service.
3. Clicked Create bucket.
4. Entered the bucket name my-music-uploads.
5. Selected the Asia Pacific (Mumbai) – ap-south-1 region.
6. Kept the default bucket settings and clicked Create bucket.
7. Opened the newly created bucket.
8. Clicked Upload.
9. Selected song1.mp3 and song2.mp3 from the local computer.
10. Clicked Upload and verified that both files were uploaded successfully.

RESULT

The S3 bucket my-music-uploads was created successfully, and both MP3 files were uploaded successfully.

EVIDENCE





02 Apply a bucket policy to 'my-music-uploads' so only users with a specific email address (your own or a friend's) can download files from the bucket.

Hint: Use the bucket's access management settings to add an IAM user or assign a permission for a specific email.

Answer: A bucket policy was applied to the my-music-uploads S3 bucket to allow only a specific IAM user to download objects from the bucket.

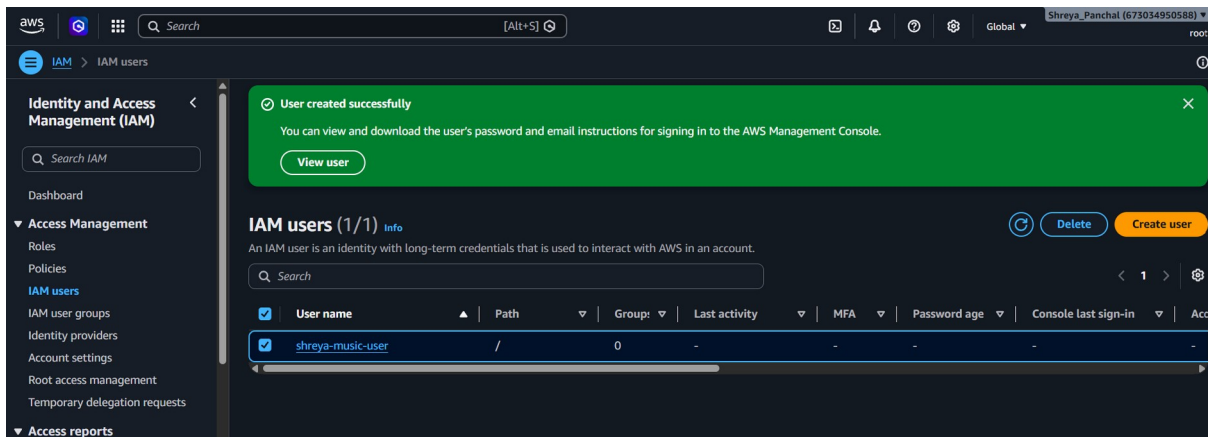
STEPS PERFORMED

1. Logged in to the AWS Management Console.
2. Opened the IAM service and created a new IAM user named music-user.
3. Copied the IAM user's ARN.
4. Opened the Amazon S3 service and selected the my-music-uploads bucket.
5. Navigated to the Permissions tab and opened Bucket Policy.
6. Added a bucket policy granting the IAM user s3: GetObject permission.
7. Saved the bucket policy successfully.

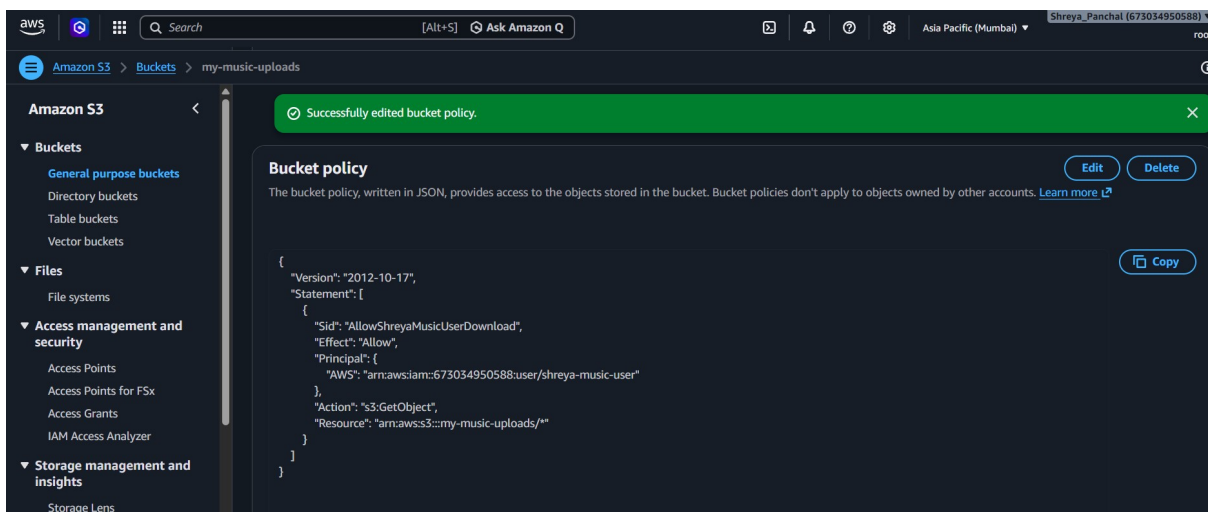
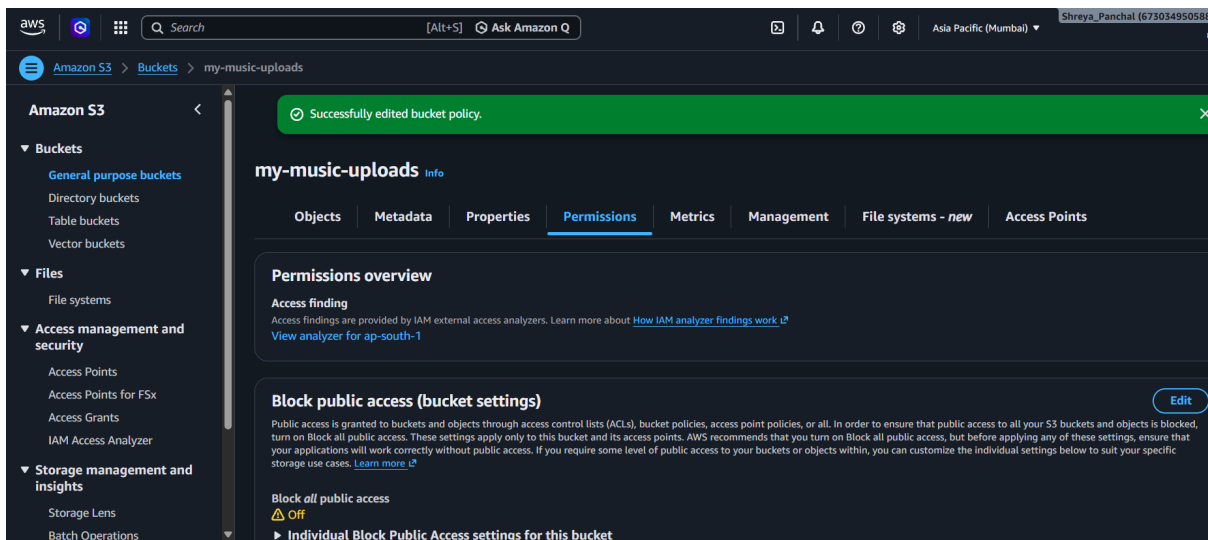
RESULT

The bucket policy was successfully configured so that only the specified IAM user has permission to download files from the S3 bucket.

EVIDENCE



IAM User ARN: `arn:aws:iam::673034950588:user/shreya-music-user`



03 Enable server-side encryption for all objects uploaded to 'my-music-uploads', ensuring that future uploads are encrypted automatically by the cloud provider.

Answer: Server-side encryption was enabled for the my-music-uploads S3 bucket to ensure that all future uploaded objects are automatically encrypted.

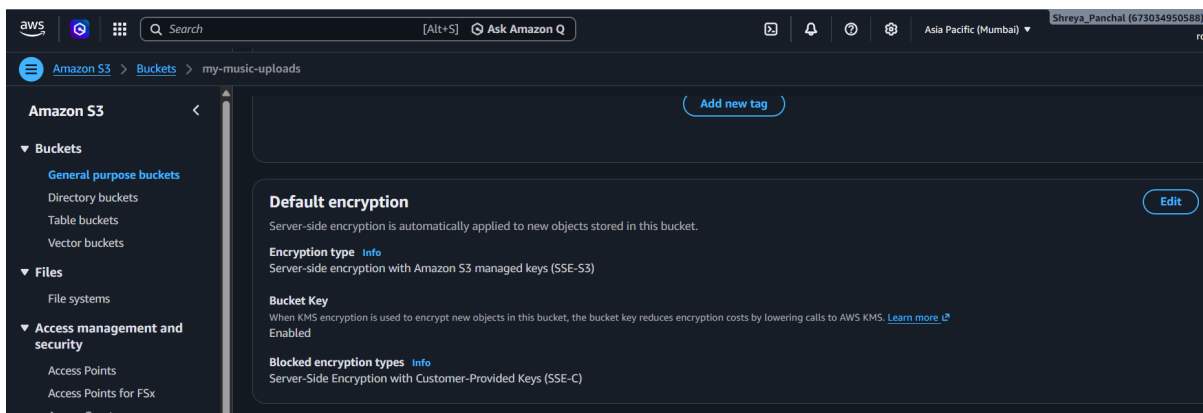
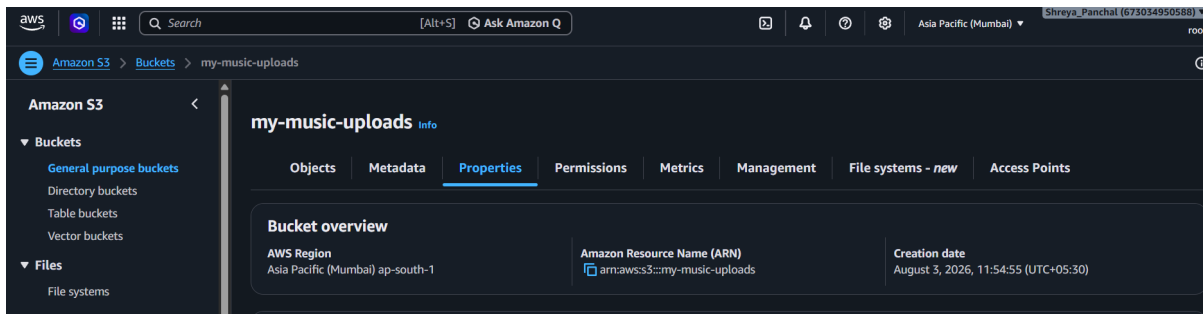
STEPS PERFORMED

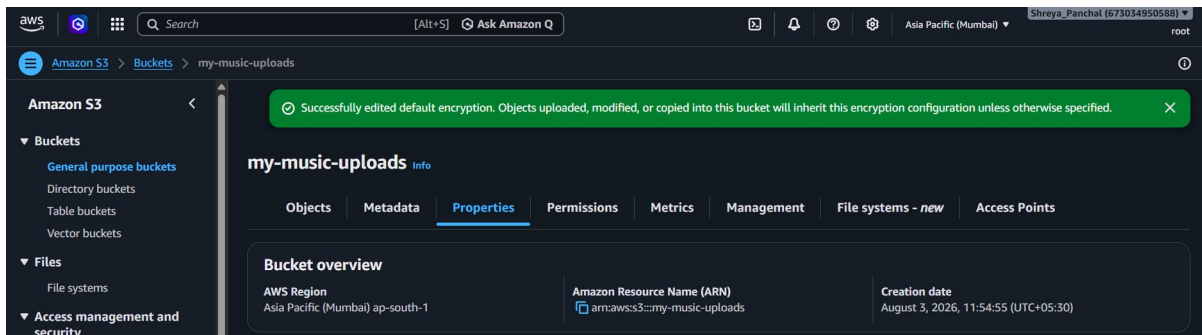
1. Logged in to the AWS Management Console.
2. Opened the Amazon S3 service.
3. Selected the my-music-uploads bucket.
4. Opened the Properties tab.
5. Navigated to the Default Encryption section and clicked Edit.
6. Selected Server-side encryption with Amazon S3 managed keys (SSE-S3).
7. Left the Bucket Key setting as Disabled.
8. Clicked Save changes to enable default encryption.

RESULT

Server-side encryption was successfully enabled for the bucket. All future objects uploaded to my-music-uploads will be automatically encrypted using Amazon S3 managed keys (SSE-S3).

EVIDENCE





04 Test your bucket's security by trying to access an uploaded MP3 file from a browser where you are not logged in — take a screenshot of the error or access denied message.

Answer: The security of the S3 bucket was tested by accessing the Object URL of an uploaded MP3 file from an Incognito (private) browser window without logging in to AWS.

STEPS PERFORMED

1. Opened the Amazon S3 bucket my-music-uploads.
2. Selected an uploaded MP3 file and copied its Object URL.
3. Opened a new Incognito/Private browser window.
4. Pasted the Object URL into the browser and attempted to access the file.
5. Verified that access was denied because the bucket policy does not allow anonymous users to download objects.

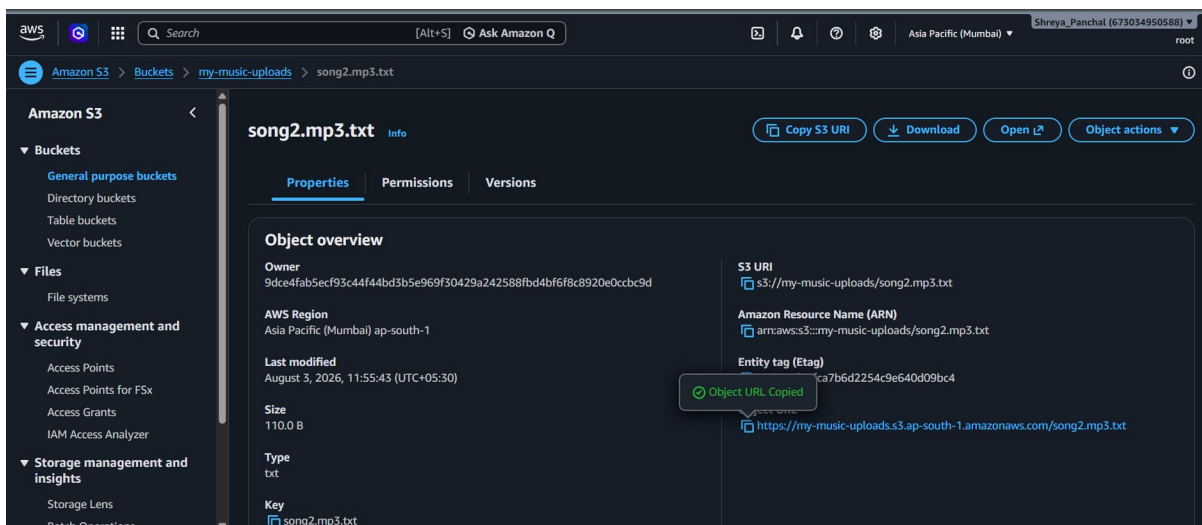
RESULT

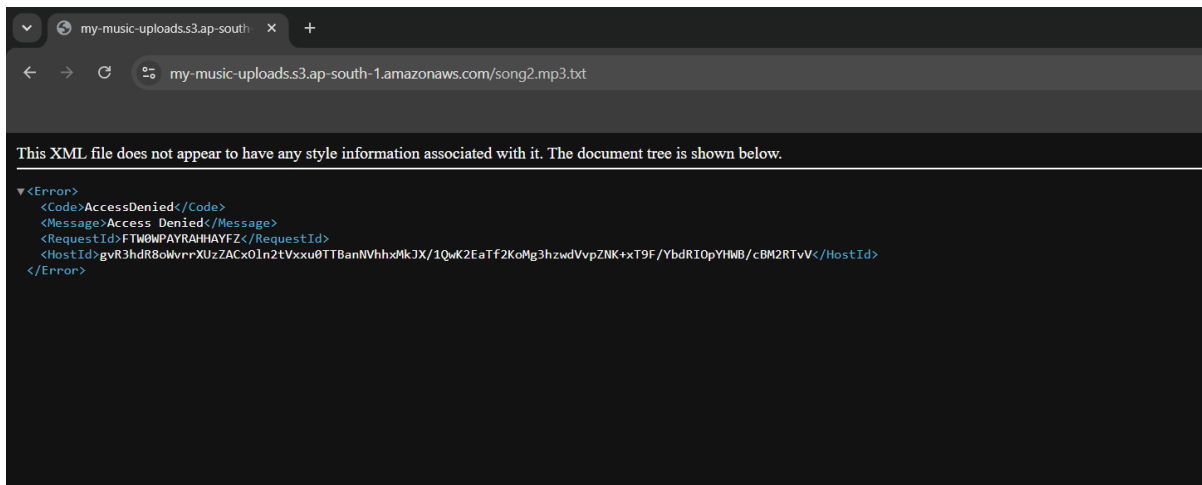
The browser displayed the following message:

Access Denied
Access Denied

This confirms that the S3 bucket is securely configured and prevents unauthorized users from accessing the uploaded files.

EVIDENCE





05 Use an AI tool like ChatGPT or Copilot to generate a sample bucket policy that restricts access to your bucket by IP address, then modify and apply it to 'my-music-uploads'.

Hint: Copy the AI-generated policy, review it for your cloud provider's syntax, and adjust the IP to your college's or your own network.

Answer: A sample S3 bucket policy was generated using ChatGPT and modified to allow access only from my public IP address 171.61.167.128/32. The policy was applied to the my-music-uploads bucket through the Permissions → Bucket Policy section in Amazon S3.

STEPS PERFORMED

1. Used ChatGPT to generate an IP-based S3 bucket policy.
2. Found my public IP address (171.61.167.128).
3. Updated the policy by replacing the sample IP with 171.61.167.128/32.
4. Opened Amazon S3 → my-music-uploads → Permissions → Bucket Policy.
5. Pasted the updated policy and clicked Save changes.
6. Tested the Object URL from my current network and verified that access was allowed.

RESULT

The bucket policy was successfully applied. Access to the bucket objects is restricted to requests originating from the specified IP address (171.61.167.128/32).

EVIDENCE

whatismyipaddress.com

Enter Keywords or IP Address... **Search** ABOUT PRESS BOOK PODCAST CONTACT

MY IP IP LOOKUP HIDE MY IP VPNS TOOLS LEARN

GoDaddy Airo Tell Me More

My IP Address is:

IPv6: **2401:4900:8fed:2bf7:487:2448:51a7:92d1**

IPv4: **171.61.167.128**

My IP Information: Your location may be exposed!

ISP: Bharti Airtel Ltd. **HIDE MY IP ADDRESS NOW**

Map showing location near Ahmedabad, Gujarat.

Amazon S3 Buckets my-music-uploads Edit bucket policy

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Bucket ARN: `arn:aws:s3:::my-music-uploads`

Policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "AllowAccessFromSpecificIP",
6       "Effect": "Allow",
7       "Principal": "*",
8       "Action": "s3:GetObject",
9       "Resource": "arn:aws:s3:::my-music-uploads/*",
10      "Condition": {
11        "IpAddress": {
12          "aws:SourceIp": "171.61.167.128/32"
13        }
14      }
15    }
16  ]
17 }
```

Edit statement: AllowAccessFromSpecificIP Remove

Add actions

Choose a service

Search services

Included: S3

Available: AI Operations

Amazon S3 Buckets my-music-uploads song1.mp3.txt

song1.mp3.txt Info

Copy S3 URI Download Open Object actions

Properties Permissions Versions

Object overview

Owner: `9dce4fab5ecf93c44f44bd3b5e969f30429a242588fbd4bf6f8c8920e0ccbc9d`

AWS Region: Asia Pacific (Mumbai) ap-south-1

Last modified: August 3, 2026, 11:55:44 (UTC+05:30)

Size: 116.0 B

Type: txt

Key: `song1.mp3.txt`

S3 URI: `s3://my-music-uploads/song1.mp3.txt`

Amazon Resource Name (ARN): `arn:aws:s3:::my-music-uploads/song1.mp3.txt`

Entity tag (Etag): `e0c6e7ee62118c08086eb2`

Object URL Copied

`https://my-music-uploads.s3.ap-south-1.amazonaws.com/song1.mp3.txt`

